

Activity - 2

1. Salt and pepper noise

Technique used: Median Filter

Reason:

Median filtering effectively removes salt and pepper noise by replacing noisy pixels with the median value of neighboring pixels. It preserves image edges better than averaging filters.

2. Gaussian noise

Technique used: Gaussian Filter

Reason:

A Gaussian filter smooths the image by reducing Gaussian noise while maintaining noisy pixels and the overall image structure. It is commonly used for sensor-generated noise.

3. Random noise

Technique used: Bilateral Filter

Reason:

Bilateral Filtering removes random noise while preserving important edges and fine anatomical details. This makes it ideal for medical imaging applications.

4. Black and white dots

Technique used: Median Filter

Reason:

Median Filter efficiently removes isolated black and white dots without blurring the document text. It improves readability while preserving character edges.

5. Heavy grain

Technique used: Gaussian Filter

Reason:

The Gaussian Filter reduces grainy noise by smoothing intensity variations. It enhances the visual appearance when edge preservation is not the primary concern.

6. Very low contrast

Technique used: Histogram Equalization

Reason:

Histogram Equalization redistributes pixel intensities to improve image contrast. It enhances the visibility of roads, vehicles and objects in foggy conditions.

7. Dull and washed out

Technique used: Contrast Stretching

Reason:

Contrast stretching expands the intensity range, making colors and details more visible. It improves the appearance of underwater images affected by poor lighting.

8. Poor visibility

Technique used: Adaptive Histogram Equalization (CLAHE)

Reason:

CLAHE enhances local contrast without over-amplifying noise. It makes bones and soft tissues more distinguishable in medical x-ray images.

9. Lacks intensity variation

Technique used: Histogram Equalization

Reason:

Histogram equalization increases the intensity range, improving brightness and visibility in low-light traffic images. It helps identify vehicles and road features.

10. Narrow intensity range

Technique used: Contrast stretching

Reason:

Contrast stretching expands the limited intensity values across the full grayscale range. This produces a clearer image with improved visual detail.